Filling 10 ms Charge-Exchange Spectroscopy Gaps in KSTAR: What Co-located Fast Diagnostics Add to the Ion-Temperature Past, and a Bound for Rotation

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Abstract

Charge-exchange spectroscopy (CES) gives KSTAR's ion temperature T_i and toroidal rotation V_{rot} at one chord ($R = 2.198$ m); on the 10 ms grid of 641 discharges of the 2022 campaign, 8.2% of T_i and 23.9% of V_{rot} samples are missing. We ask what beam-emission spectroscopy (BES) and electron-cyclotron-emission imaging (ECEI) channels within 1.5 cm of that chord add to the CES past when a step must be filled without the next CES measurement. A two-branch LSTM reads the fast channels (standardized with each discharge's own statistics, an offline step) and the CES past for T_i, and only the CES past for V_{rot}. For T_i, the mean-squared-error skill of this full-input causal estimator over a CES-only causal Gaussian-process (GP) smoother is +0.113 [+0.085, +0.132] (cut; 95% interval, eight re-partitions of 639 discharges; the inclusive interval also excludes zero); the fast-channel share of it is not isolated. Over PCHIP interpolation the skill is +0.221 [+0.188, +0.248], about half of which a running mean of past CES already provides; against an offline GP it is −0.010 [−0.043, +0.014]. The gain sits in the tail: the median absolute error (47.2 eV) is close to persistence's (46.1 eV) and above PCHIP's (39.4 eV; median-AE skill −0.198 [−0.259, −0.135]). A 20-parameter surrogate filter summarizes the T_i branch post hoc. For V_{rot}, the noise-corrected correlation of fast-channel and V_{rot} increments is bounded by $|r| \le 0.044$ (lags 0–200 ms, windows 10–200 ms; the T_i lower bound is at least +0.127 at lag 0 for every window). The history-only V_{rot} estimate has skill +0.265 [+0.198, +0.326] over persistence under a rule set in review that drops the 0.05% of rows whose target or anchor exceeds 1,000 km/s; its margin over the causal GP holds in a pooled readout only, and a margin over PCHIP is not established. A causally standardized variant is untested. One device, one campaign, one chord.

1. Introduction

Ion temperature T_i and toroidal rotation V_{rot} are core kinetic quantities of a tokamak plasma; here both come from one outboard chord at $R = 2.198$ m, with r/a not mapped. Both are measured by charge-exchange spectroscopy (CES) [1], which infers T_i from Doppler broadening and V_{rot} from the Doppler shift of the charge-exchange line. CES must integrate photons, so it is slow and intermittently missing: on the KSTAR 10 ms grid, 8.2% of T_i and 23.9% of V_{rot} observations are missing—independently of each other—and a further 41.1% of V_{rot} rows repeat the previous reading bit-for-bit, so only 35.0% of the grid carries an independent V_{rot} measurement. The missing rate rises from 0.77% to 4.45% across quintiles of neighbouring-step change and is 19.9% in the lowest BES-intensity quintile against 3.7% in the highest, i.e. it is missing-not-at-random [2]. Meanwhile beam-emission spectroscopy (BES), electron-cyclotron-emission imaging (ECEI) and Mirnov coils observe the same plasma without gaps, but none measures T_i or V_{rot}. We ask: *at a 10 ms step where CES is missing, what do the simultaneous fast diagnostics add to the irregular CES past?* The use case is an estimate that cannot wait for the next CES exposure, with fast channels standardized per discharge (offline); a causally standardized variant is untested under this protocol. Against an offline noise-aware smoother that reads both neighbours, the skill interval includes zero (§6.2).

Comparators. Deployable comparators read only the CES past: persistence, a running mean, and a causal Gaussian process (GP) on the 16 past CES values. Linear and PCHIP [3] interpolation and an offline GP read both neighbours and serve as offline references. PCHIP passes exactly through noisy anchors, so a smoother of the CES past alone already gains on it (running mean +0.05 to +0.17 over PCHIP, cut, interval excluding zero on 3/4 splits); the deciding deployable comparison is therefore the causal GP.

Every result is reported in two co-primary populations: *cut*, in which the 0.53% of T_i rows above 3 keV (spectral-fit failures) are treated as missing in every arm, and *inclusive*, in which they are kept (§3); the inclusive population's extra margin is mostly spike handling.

Claims. *(1) Ion temperature.* The full-input causal estimator exceeds the CES-only causal GP by +0.113 [+0.085, +0.132] (cut; +0.118 [+0.076, +0.157] inclusive); the fast-channel share is not isolated by a retrained history-only model. *(2) Rotation.* The same channels' noise-corrected correlation with V_{rot} increments is bounded by $|r| \le 0.044$ at 10–200 ms. V_{rot} is estimated from its own history: skill over persistence, a copy-last baseline, +0.265 [+0.198, +0.326] (cut, under the fit-failure rule set in review); over the causal GP +0.151 [+0.038, +0.290] pooled, per split 1/4 (cut). Its median row is worse than persistence (4.3 vs. 3.7 km/s), and its 10 ms scatter is consistent with Gaussian line-fit errors ($\sqrt{2}$ ratio; $\sqrt{T_i}$ scaling not observed). The evaluation (two co-primary populations, a shot-clustered paired bootstrap, an overlap-corrected repeated

cross-validation interval and a two-level bootstrap, §5) is assembled from standard pieces and is described as method, not claimed. Code and evaluation artifacts are released; the KSTAR shot files are not redistributable.

2. Related work

Gap filling and baselines. GP regression [4] is the standard tool for fusion profile fitting [5, 6], applied to KSTAR CES T_i profiles [7]; PCHIP [3] avoids spline ringing at steps. We score with Murphy's skill [8]. Persistence as a reference has precedent in fusion profile forecasting [9] and space weather [10].

Neural diagnostic inference. NN-CES extracts T_i and rotation from KSTAR CES spectra without beam modulation [11]; it is the closest prior work. Rows with no CES exposure belong to this work; rows with an existing spectrum but a rejected fit belong to spectral re-analysis such as NN-CES. The files do not record which case applies, so the split between the two is not measured; per-exposure fit metadata would resolve it and would replace the value cut. Lin et al. [12] map an x-ray crystal spectrometer to T_i and rotation on EAST, an instantaneous mapping from the same photon channel. RTCAKENN [13] reconstructs kinetic profiles in real time and tolerates dropped Thomson/CER inputs; there missingness is a robustness condition, here the target's own sparsity is the problem.

Cross-diagnostic reconstruction. Diag2Diag synthesizes Thomson T_e/n_e from simultaneous diagnostics on DIII-D [14] with a memory-less MLP; related work densifies Thomson on COMPASS [15], fills Thomson intervals causally with an extended Kalman filter on TCV [16], restores EAST T_e and Z_{eff} [17, 18] and masks channels on HL-3 [19]. These target electron or generic dense channels; we extend the programme to a sparse ion channel, to estimation conditioned on the target's own irregular past, and to per-target tests against pre-registered baselines.

Methods lineage. Observation masks follow GRU-D [20]; the overlap-corrected interval is [21], the repeat-then-discharge resampling is a hierarchical bootstrap [22], and the surrogate retention test follows the fixed-point readout of [23].

3. Data and problem

We use 641 KSTAR discharge files (shots 30801–32751) aligned to a 10 ms grid, 247,207 rows in total. Each row holds BES [24] (9 channels), ECEI [25] (4), Mirnov coils (2), time, and the two CES targets. T_i is missing in 8.2% of rows and V_{rot} in 23.9%, independently, so the loss is masked per target (Fig. 1). Both CES targets are the 2022 channel at $R = 2.198$ m; the nine BES and four ECEI channels lie within about 1.5 cm of it, and the two Mirnov inputs are two positions of one poloidal array at one toroidal location.

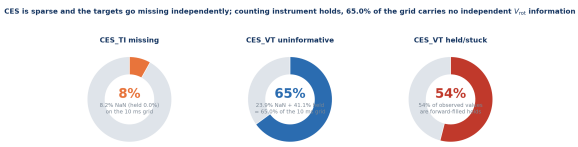


Figure 1: CES missingness on the 10 ms grid. T_i and V_{rot} go missing independently (8.2% / 23.9%); counting holds, 65.0% of the grid carries no independent V_{rot} information.

Rows group into contiguous segments at gaps ≥ 0.5 s; no arm bridges a gap.

Held V_{rot} values. 54% of observed V_{rot} values are bit-identical repeats of the previous observation (runs up to 1,214 rows, 499/641 files); the files do not record why; the T_i and V_{rot} masks agree on 72.8% of rows, so the two channels are post-processed separately. T_i has 1 held row in 226,991. Holds are removed *everywhere*—targets, history inputs, normalization statistics and every baseline's anchors—so no arm scores a forward-fill for free.

Fit failures and two populations. 1,197 T_i rows (0.53%) exceed 3 keV (951 runs in 274 discharges, 85% of runs single rows), failed spectral fits that also poison interpolation anchors. We pre-registered **two co-primary populations**: *cut* (>3 keV treated as missing in every arm) and *inclusive* (no cut); a claim is stated without qualification only if it holds in both. The threshold is immaterial across 2.5–4 keV (mean T_i skill +0.230 and +0.232 against +0.236 at 3 keV).

4. Model

The main model, `seq_v2`, is a 357,570-parameter sequence nowcaster. This section fixes what is estimated, why the inputs are wired to each target, which time scales the model sits on, and a 20-parameter surrogate of its T_i branch.

4.1. Estimand, loss and an irreducible floor

At grid time t the model estimates the normalized $y_t = [T_i, V_{\text{rot}}]_t$ from $x_{\le t}$ with per-target observation masks $m_{t,c} \in \{0, 1\}$:

$$\mathcal{L} = \frac{\sum_{t,c} m_{t,c}\,(\hat{y}_{t,c} - y_{t,c})^2}{\sum_{t,c} m_{t,c}} + \lambda\,\overline{\mathrm{ReLU}(z_0 - \hat{y}_{t,T_i})}, \qquad \lambda = 0.1, \quad (1)$$

whose first-term minimizer is $\mathbb{E}[y_t \mid x_{\le t}]$; the second term discourages T_i below the normalized zero z_0. A row with one observed target still supervises it (a both-required filter would discard ≈28% of labelled rows). The target is itself a photon-integrated fit with noise σ_{meas}, so even a perfect physical estimate scores $\text{RMSE}^2 = \sigma^2_{\text{meas}} + b^2$. Difference estimators put σ_{meas} for T_i (cut) at 46.4–129.9 eV, the order of the backbone's 157.8 eV, and 46.6% of the fourth-difference squared mass (65.6% for V_{rot}) sits in the top 1% of one-step changes: (1) is a *tail statistic* (§6.5).

4.2. Two conservation laws and the input routing

Ion energy and toroidal angular momentum obey

$$\partial_t\left(\tfrac{3}{2}n_iT_i\right) + \nabla\cdot\mathbf{q}_i = Q_{ei}\,(T_e - T_i) + S_i, \tag{2}$$

$$\partial_t\left(n_i m_i \langle R^2\rangle \omega_\varphi\right) + \nabla\cdot\Pi_\varphi = T_{\mathrm{NBI}} + T_{\mathrm{NTV}} + T_{\mathrm{int}} + \cdots \tag{3}$$

Both equations constrain a rate. Of the terms in (3), no torque, beam-power or NTV channel exists here (Mirnov dB/dt cannot see a static δB), BES intensity is not the momentum flux $\langle v_r v_\varphi\rangle$, and residual stress needs ∇T_i, so little V_{rot} information is expected from the fast channels; the measured bound is in §6.3. Under an earlier protocol (before the fit-failure cut), a full-grid variant with a shared encoder had paired V_{rot} skill −0.044, −0.072, −0.067, −0.023 against the windowed control (4/4 excluding zero); with branches split at the encoder and per-discharge standardization it had −0.026, −0.003, −0.039, −0.013 (0/4). No paired shared-vs-split comparison exists under the present protocol, and these values predate the V_{rot} fit-failure rule. For T_i the fast-channel response is a same-instant co-movement of channels within 1.5 cm of the chord (§4.5); its mechanism is not identified here, since per-discharge standardization removes the absolute T_e and n_e that collisional exchange would need and the files carry no beam-power or D-alpha signal.

4.3. Time scales

The grid spacing $\Delta t = 10$ ms gives $f_{\mathrm{Nyq}} = 1/(2\Delta t) = 50$ Hz. Mode rotation sits in the kHz band (edge harmonic oscillations near 4 and 8 kHz in this dataset), so a dB/dt stream decimated to 100 Hz without anti-aliasing cannot carry it; the Mirnov lag-1 autocorrelation of −0.009 (BES +0.568) confirms it. Autocorrelation $1/e$ times agree for T_i (159 ms), BES (161) and ECEI (147), whereas V_{rot} gives 16 ms without holds and more than 300 ms with them: its relaxation time cannot be measured in this dataset.

4.4. Full-grid sequence backbone

`seq_v2` keeps every input-complete row of a contiguous segment, labelled or not, as context, because the fast diagnostics have no gaps (Fig. 2). Each step carries 22 channels; every CES-derived input uses rows before t only, while the 15 fast channels are standardized with the discharge's own mean and variance, an offline step. The CES-derived channels are $\log(1+\Delta t_{\mathrm{prev}})$ and per target the last observed value carried forward from rows $< t$, log(1+staleness) and a has-any-past-observation flag. The T_i branch reads all of them; the V_{rot} branch reads only the seven non-fast channels—the implementation of §4.2. Because the split is at the encoder, no shared recurrent state carries fast information into the V_{rot} head. Outputs are normalized $[T_i, V_{\mathrm{rot}}]$ per row, trained with (1) over every labelled row (AdamW, $\eta = 10^{-3}$, weight decay 10^{-4}, batch 16 segments, clipping 1.0, early stopping with a 30-epoch cap). Splits are file-level, normalization of every other channel is train-file-only, and the target row never enters the input. Under this protocol the cost of a causal (training-statistics) standardization is not measured; under an earlier protocol without the fit-failure cut, that variant kept a T_i skill over PCHIP of +0.22 to +0.30 with intervals excluding zero on 4/4 splits. Truncating the backbone's own context costs −0.066 at 20 ms and saturates at 50 ms: the past reaches the T_i output through the carried CES value and about 50 ms of fast-channel context.

Windowed control and the gate. The control (201,258 parameters), which runs without per-discharge standardization, encodes each modality with its own time-aware 1-D CNN over the W=2 preceding rows and reads the CES history with a bidirectional GRU pooled by one multi-head attention readout per target, hard-masked so that attention mass lands only on rows where that target was measured; its V_{rot} head also fuses history and time only. The full-grid framing was adopted over it by a pre-registered four-condition gate on a 4×4 grid of split × initialisation seeds: paired T_i skill positive on 16/16 runs, pooled +0.081 [+0.067, +0.096]. The control's skill over the causal GP is +0.041 (interval including zero on 3/4 splits, cut).

Cost. A stateful step costs 0.97 ms median and 1.49 ms at p99 on CPU; deployment needs causal standardization.

4.5. A 20-parameter surrogate of the T_i branch

A pre-registered readout of the thirty frozen `seq_v2` checkpoints found that weight magnitude carries no stable information (first-layer column norms are flat across the fifteen fast channels and per-channel ranks do not reproduce across splits), whereas the input–output response does: the T_i output is an exponentially weighted average of past observed T_i plus a same-instant response to the fast channels, with a forget-gate memory of 1.3 grid steps. A frozen-row readout passed its registered bar, but a first trained increment form fell below the registered 0.70 bar; the backbone's Jacobian on the test rows then suggested Eq. (4): a ≈100 ms high-pass fast-channel response (current value minus the preceding ten-step mean) with gain flat to 2σ and switched off beyond 3σ. Because it was read on the test rows of the same four splits on which the form is scored, the retention is post hoc. The corrected form, trained from scratch under the identical protocol, is

$$\hat{y}_t = m_t + \sum_{c=1}^{15} a_c\, d_{c,t}\,\sigma\!\left(\frac{\theta - |d_{c,t}|}{0.5}\right) + b_0, \tag{4}$$

$$d_{c,t} = x_{c,t} - \bar{x}_{c,\,t-10..t-1},$$

$$m_t = m_{t-1} + K(s)\,(y^{\mathrm{obs}} - m_{t-1}),$$

$$K(s) = K_\infty + (K_0 - K_\infty)\,e^{-s/s_K}, \tag{5}$$

where m_t is updated only when a new T_i measurement arrives, s is the gap since the previous one, and $\sigma(\cdot)$ is the logistic function; $d_{c,t}$ and θ are in per-discharge standard deviations

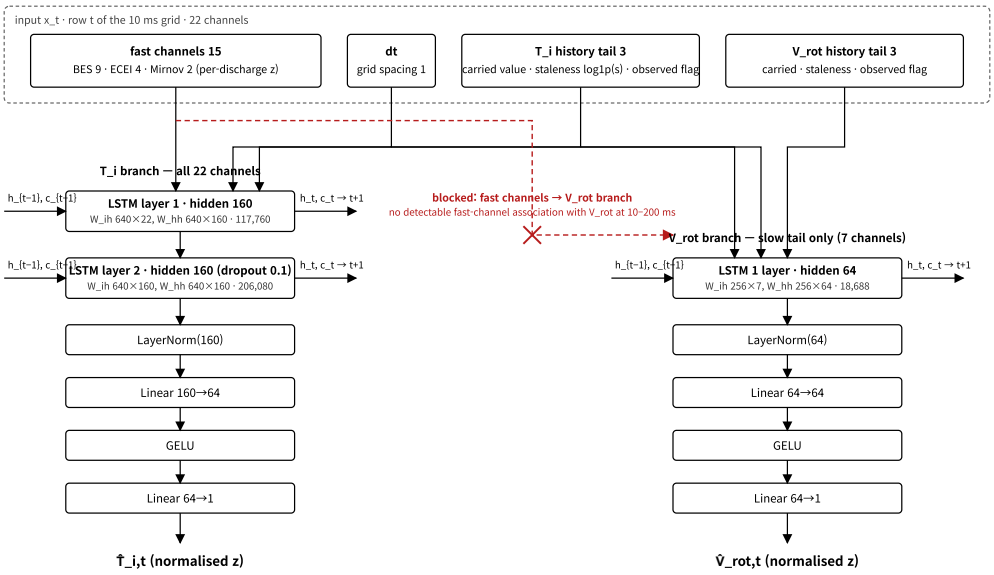


Figure 2: `seq_v2` for one grid row t. 22 inputs (15 fast, per-discharge z (whole-discharge statistics); Δt; per target carried value, staleness, observed flag) feed two LSTMs that run forward in time. The T_i branch reads all 22 (2 layers, 160 units), the V_{rot} branch only the seven slow channels (1 layer, 64 units); each ends in LayerNorm and Linear–GELU–Linear. The dashed red path from the fast channels to V_{rot} does not exist by construction. 357,570 parameters.

(offline normalization). The first observation of a segment is taken whole ($K = 1$). Its 20 parameters are three gain constants, one gain a_c per fast channel, the threshold θ and the offset b_0. Across the four cut splits the learned values are $K_0 = 0.27$–0.30 for a fresh measurement, $K_\infty = 0.48$–0.53 after a long gap, transition $s_K = 39$–47 ms, $\theta = 2.9$–3.0σ, $\sum a_{\mathrm{BES}} = +48$ to $+53$ and $\sum a_{\mathrm{ECEI}} = +44$ to $+50$ eV per σ. Read as a filter: a Kalman-like observation update that trusts a new CES sample more the longer it has waited, plus a linear response to each fast channel's departure from its 100 ms baseline that rejects outliers beyond 3σ. Since $d_{c,t}$ is invariant to the per-discharge mean, the surrogate's offline dependence is through the per-discharge σ only. Its retention is in §6.4.

5. Evaluation protocol

Metric and baselines. Errors are computed per target in physical units, and the primary score is $\mathrm{skill}_{\mathrm{PCHIP}} = 1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{PCHIP}}$ (0 = equal error). Every arm is scored on the identical (file, row) set. Persistence, a running mean and two-point extrapolation read only the past; linear, PCHIP and an offline GP (Matérn-3/2 + white noise, nearest 16+16 neighbours, per-sample marginal-likelihood selection) read the future; the causal GP is the same GP on the 16 past neighbours. Where no future neighbour exists the interpolant predicts persistence (0.2% of scored T_i rows and 0.3–0.5% of scored V_{rot} rows).

Splits and pre-registration. Files are split train/val/test with four seeds (42, 1, 7, 123). Test files were reserved before the architecture search, but the adoption of the sequence backbone, of per-discharge standardization and the surrogate's form used results on the same four test splits; the repeated cross-validation re-partitions the same 639 discharges. The canonical test set holds 32,589 T_i rows over 96 shots and 10,463 V_{rot} rows over 60 shots (cut). "Pre-registered" means the hypotheses, decision rules and numerical predictions of a batch were committed to the repository before it was run; there is no third-party registry.

Intervals. Per-sample paired squared-error differences are aggregated by shot and whole shots resampled (10,000 resamples); a result is reported with whether its 95% interval excludes zero. With four splits, each split's value is given with its own interval, together with the range and the number of splits whose interval excludes zero, rather than a mean ± standard deviation of four numbers [26, 27]. Because the four splits share most of their training discharges, the backbone was re-run as shot-level repeated 5-fold cross-validation on 639 discharges (ten repeats, 100 runs, every discharge tested once per repeat). *Fold-level*: the Nadeau–Bengio interval $\bar{S} \pm t_{0.975,n-1}\sqrt{(1/n + n_{\mathrm{test}}/n_{\mathrm{train}})\, s^2}$ [21]. *Pooled*: each repeat's five test folds concatenated into one skill, with a two-level bootstrap that draws a repeat and then resamples its discharges. The repeated cross-validation was registered with the fold-level interval as its deciding readout; for V_{rot}

Table 1: RMSE, seed-42 test split, cut population, genuine measurements only (n = 32,589 T_i, 10,463 V_{rot}). Blocks: past CES only; past CES + fast channels, standardized per discharge (offline step; the windowed control uses training statistics); past + future CES. Inclusive: `seq_v2` 363.0 / 23.7, PCHIP 412.4 / 30.2, causal GP 394.6 / 28.8, persistence 478.0 / 33.4. V_{rot} RMSEs are without the fit-failure rule of §6.2; under it (2 rows dropped on this split) the backbone's V_{rot} skill is +0.088 [+0.019, +0.145] over the causal GP, +0.215 [+0.071, +0.289] over PCHIP and +0.329 [+0.265, +0.368] over persistence.

Arm	Information	T_i (eV)	V_{rot} (km/s)
Causal GP	past 16 CES	164.3	28.8
Running mean	past CES	169.2	–
Persistence	last observed CES	197.2	33.4
Two-point extrap.	past 2 CES	472.2	51.0
`seq_v2`	fast + past CES, segment	**157.8**	**23.6**
20-param. surrogate[a]	fast + past CES	162.3	–
Windowed control	fast + 2 rows past CES	169.2	26.1
GP (offline)	past + future CES	153.8	24.7
Linear interp.	past + future CES	169.8	29.0
PCHIP	past + future CES	173.6	30.2

[a] An interpretation of the backbone, not a standalone-estimator claim; paired skill over the causal GP on this split +0.024 [−0.012, +0.057].

against PCHIP it included zero. A pooled readout was then computed post hoc, and eight new repeats were registered with the pooled two-level readout as deciding before any of them ran. That interval conditions on the trained models and does not carry the correlation between overlapping training sets; the fold-level interval does.

6. Results

6.1. RMSE ladder

Table 1 gives the physical-unit RMSE on the canonical test split. Among the methods that do not read future CES values, `seq_v2` has the lowest RMSE, 4% (T_i) below the causal GP, and it is also below linear and PCHIP interpolation. The offline GP, which averages noisy anchors, is the one arm with a lower T_i RMSE (153.8 vs. 157.8). Linear interpolation is the stronger offline interpolant here: per split, the T_i skill over it is +0.136 to +0.226 (cut) and +0.209 to +0.288 (inclusive), the interval excluding zero on 4/4 splits in both populations, 0.01–0.05 below the skill over PCHIP; for V_{rot} it excludes zero on 1/4 splits in each population.

6.2. Headline and replicate

T_i. Table 2: $\mathrm{skill}_{\mathrm{PCHIP}}$ = +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive), every shot-clustered interval excluding zero on all four splits; over the causal GP +0.08 to +0.17 and over persistence +0.36 to +0.46, both excluding zero on 8/8 population–split cells. Against the offline GP the paired skill is −0.05 to +0.11 with the interval including zero on 7/8 cells: the margin is over interpolants and causal smoothers, not over every offline smoother. The replicate reproduces every verdict: vs. PCHIP the fold-level interval is [+0.19, +0.25] (cut) and [+0.19, +0.31] (inclusive). The skill is not confined to adjacent history: beyond $\Delta t > 15$ ms since the last observation it is +0.268 (cut) and +0.206 (inclusive) over PCHIP, both intervals excluding zero.

V_{rot}. Point estimates are positive on every split, but the four-split interval over PCHIP excludes zero on 1/4 (cut) and 2/4 (inclusive) splits, and the fold-level interval includes zero ([−0.02, +0.41] cut). The pooled readout over eight re-partitions of the same 639 discharges excludes zero over PCHIP, persistence and the causal GP (Table 2), but that margin over PCHIP rests on fit failures. A rule that removes, from every arm, the rows whose target, persistence anchor or PCHIP anchor exceeds $|V_{\mathrm{rot}}| = 1{,}000$ km/s drops 36–41 of about 71,700 rows per repeat (0.05%), which carry 63–78% of PCHIP's squared error (cut). This rule was set during review, after the share carried by these rows was known; it is applied when scoring only, so the history inputs still contain the fit failures and no model was retrained. Under this rule the pooled skill over PCHIP is +0.092 [+0.0003, +0.170] (cut) and +0.090 [−0.010, +0.177] (inclusive), and it includes zero in both populations at thresholds of 500 and 2,000 km/s, so a V_{rot} margin over PCHIP is not established. Over persistence it is +0.265 [+0.198, +0.326] (cut) and +0.257 [+0.183, +0.323] (inclusive), per split 4/4 in both populations at 500 and 1,000 km/s and 3/4 at 2,000 km/s. Over the causal GP it is +0.151 [+0.038, +0.290] and +0.132 [+0.026, +0.267], excluding zero at all three thresholds; per split it is +0.057 to +0.131 (cut, interval excluding zero on 1/4) and +0.053 to +0.143 (inclusive, 2/4), every point estimate positive; a fold-level interval under the rule was not computed. Against the offline GP the history-only estimate has −0.073 [−0.162, +0.005] (cut, including zero) and −0.087 [−0.174, −0.008] (inclusive). The V_{rot} margin is established against persistence on every split; against the causal GP only in the pooled readout.

Why inclusive numbers are higher. Every arm scores higher against PCHIP without the cut (backbone mean +0.268 vs. +0.236) because spikes poison the interpolation anchors, which a learned model can discount; this is why neither population is quoted alone.

Two stress tests. *Missing-not-at-random:* reweighting the scored rows onto the covariate distribution ($\Delta t \times$ local activity) of the genuinely missing, in-domain rows—53.6–67.0% (cut) and 54.1–67.7% (inclusive) of missing T_i rows—gives T_i skill over PCHIP +0.14 to +0.28 with intervals excluding zero on 2/4 cut and 4/4 inclusive splits, and over persistence +0.28 to +0.44 with intervals excluding zero on 8/8 cells (missing V_{rot} rows are only 4–6% in-domain, and we draw nothing from them). *Campaign shift:* with a strictly temporal split (train 416 shots, val 128, test the 97 latest), `seq_v2`

Table 2: Skill of `seq_v2` against each comparator. *Four splits*: range of the per-split skill over seeds 42/1/7/123 with the number of splits whose shot-clustered 95% CI excludes zero. *Fold-level*: Nadeau–Bengio 95% interval over 50 folds of repeated 5-fold CV, with the number of folds whose own CI excludes zero; this was the deciding readout as first registered. *Pooled*: median and two-level bootstrap 95% interval over eight re-partitions of the same 639 discharges, registered as deciding before those repeats ran. Bold = interval excludes zero. The four-split offline GP entries pool both populations. V_{rot} rows are the registered readout without the fit-failure rule of §6.2.

		Four splits		Fold-level (repeated 5-fold)		Pooled (two-level)	
Target	vs.	cut	inclusive	cut	inclusive	cut	inclusive
T_i	PCHIP	+0.174 to +0.264 (**4/4**)	+0.225 to +0.316 (**4/4**)	[**+0.186**, **+0.253**] (50/50)	[**+0.188**, **+0.307**] (46/50)	+0.221 [**+0.188**, **+0.248**]	+0.248 [**+0.192**, **+0.294**]
	persistence	+0.360 to +0.415 (**4/4**)	+0.406 to +0.459 (**4/4**)	[**+0.368**, **+0.410**] (50/50)	[**+0.361**, **+0.445**] (50/50)	+0.391 [**+0.369**, **+0.408**]	+0.404 [**+0.364**, **+0.438**]
	causal GP	+0.078 to +0.138 (**4/4**)	+0.123 to +0.169 (**4/4**)	[**+0.084**, **+0.138**] (47/50)	[**+0.067**, **+0.170**] (40/50)	+0.113 [**+0.085**, **+0.132**]	+0.118 [**+0.076**, **+0.157**]
	offline GP	−0.05 to +0.11 (CI includes 0 on 7/8)		[−0.044, +0.020] (1/50)	[−0.026, +0.094] (12/50)	−0.010 [−0.043, +0.014]	+0.035 [−0.017, +0.075]
V_{rot}	PCHIP	+0.135 to +0.390 (1/4)	+0.132 to +0.384 (2/4)	[−0.021, +0.405] (15/50)	[−0.053, +0.391] (11/50)	+0.313 [**+0.072**, **+0.701**]	+0.319 [**+0.046**, **+0.683**]
	persistence	+0.30 to +0.50 (3/4 in each)		[**+0.190**, **+0.601**] (42/50)	[**+0.159**, **+0.587**] (38/50)	+0.451 [**+0.277**, **+0.763**]	+0.451 [**+0.139**, **+0.747**]
	causal GP	+0.020 to +0.331 (2/4)	+0.018 to +0.324 (2/4)	[−0.060, +0.512] (11/50)	[−0.096, +0.502] (10/50)	+0.361 [**+0.045**, **+0.701**]	+0.357 [**+0.023**, **+0.683**]
	offline GP	−0.07 to +0.09 (CI includes 0 on 7/8)		[−0.234, +0.312] (0/50)	[−0.268, +0.300] (0/50)	+0.262 [−0.119, +0.635]	+0.251 [−0.139, +0.609]

keeps +0.17 to +0.20 over PCHIP and +0.11 to +0.16 over the causal GP, intervals excluding zero on 4/4 initialisations in both populations, where the windowed control keeps −0.001 to +0.091 (2/4 cut, 0/4 inclusive); for V_{rot} the skill is +0.19 to +0.23 over PCHIP (intervals including zero on 4/4) and +0.33 over persistence (4/4, cut). The fast diagnostics drift 1.22σ (BES) and 0.53σ (ECEI) between periods against 0.115σ for the targets. Per-discharge standardization was adopted after the windowed control failed this split, and it uses whole-discharge statistics, so this is not an independent test of a deployable estimator.

6.3. The $T_i \leftrightarrow V_{\mathrm{rot}}$ asymmetry

Corrected for target noise, the correlation of fast-channel and V_{rot} increments has a 95% bound $|r| \le 0.044$ over every lag (0–200 ms) and window (10–200 ms), against a T_i lower bound of at least +0.127 at lag 0 for every window 10–200 ms; this bounds a linear association and does not prove its absence (BES at 100–200 ms gives $r \approx +0.01$ with an interval excluding zero). Zeroing one input group at evaluation time on the windowed control (Table 3) feeds inputs the control never saw; the No-fast cut row (−0.125) lies below a CES-only running mean (+0.05 to +0.17 over PCHIP), so the table measures sensitivity to out-of-distribution inputs, not a group's contribution. A retrained history-only backbone has not been run. A validation-tuned running mean sits at −0.060 to +0.003 against the causal GP, so the backbone's margin over the causal GP is close to its margin over that running mean (+0.113 to +0.137, cut, 4/4; inclusive +0.061 to +0.140, 4/4); event rows concentrate it (top-10% $|d\mathrm{ECEI}|$ rows, seeds 42/1: backbone +0.116/+0.269 over PCHIP against −0.011/+0.088 for the running mean plus a linear fast-channel term). In the inclusive population the windowed control without fast channels still has +0.15 to +0.23 over PCHIP, a spike-robustness component. For V_{rot} the control's output is bit-identical without the fast channels, a consequence of the routing; the evidence for the routing is §4.2, the bound above and the two routes below.

Table 3: Evaluation-time input sensitivity of the windowed control (no retraining): $\mathrm{skill}_{\mathrm{PCHIP}}$ for T_i (4-split mean) and the paired change against the full model (* = significant splits). Zeroed inputs are out of distribution for the control, so the rows measure sensitivity, not a group's contribution.

Inputs	cut	inclusive
Full	+0.173	+0.238
No fast	−0.125; −0.25 to −0.42 (4/4*)	+0.201; −0.03 to −0.09 (2/4*)
No history	−2.11	−1.16

Two routes to rotation information. *Mirnov:* on 24 of 24 coils checked against the raw 2 MHz stream, the 100 Hz value equals a single raw sample at the grid time (lag 0), a point snapshot, so the kHz mode rotation is aliased away. *Beam torque:* no torque channel exists, so beam torque is untested here. The statement is scoped to standard CES at 10 ms at the $R = 2.198$ m chord; edge rotation inside an ELM cycle [28] is a different measurement.

6.4. Surrogate retention

Trained from scratch under the identical protocol, the 20-parameter surrogate filter (4)–(5) keeps **72–78%** of the backbone's PCHIP-relative skill and 87–90% of its persistence-relative skill on 4/4 cut splits, above the registered bar of 0.70; measured against the causal GP it keeps 0.31 to 0.50 of the backbone's skill (cut, point estimates). It sits above the first increment form by a paired +0.034 to +0.042 with intervals above zero on 4/4 splits. Because the Jacobian was read on the same test rows, this retention is post hoc. The remaining 22–28% is the network's nonlinear event handling, which the three forms tried here do not reproduce. As a standalone estimator (cut) the surrogate has skill +0.13 to +0.21 over PCHIP (4/4 splits), +0.02 to +0.07 over the causal GP (interval excluding zero on 2/4), and is below the offline GP on 4/4. These are measurements of the model; the headline numbers remain the backbone's. The bands are read from the cut population only: in the inclusive population the surrogate keeps 0.43–0.81 of the PCHIP-relative skill and its BES gain changes sign under the fit-failure spikes.

6.5. Mean and median: a dual readout

An MSE skill is decided by few rows: the backbone's top 1% of T_i squared-error rows carry 48% (cut) and 83% (inclusive) of the total. On a typical row the backbone carries its anchor: its median absolute T_i error of 47.2 eV is close to persistence's 46.1 eV and the target's own 10 ms scatter of 46.4 eV, while PCHIP reaches 39.4 eV by averaging with the future neighbour (median-AE skill −0.198 [−0.259, −0.135], win rate 0.48); the error ratio model/PCHIP falls below 1 only in the upper error quartile. For V_{rot} the backbone's median row is worse than persistence (4.3 vs. 3.7 km/s; median-AE skill vs. PCHIP −0.488 cut and −0.518 inclusive, interval below zero on 4/4 splits): its aggregate skill is a bulk deficit offset by a tail gain. Under a symmetric 1% trim and a Huber loss at the noise bound the T_i skill over PCHIP stays +0.210 and +0.156 (cut, both intervals excluding zero); the inclusive trimmed skill is +0.057 (interval including zero). A separate, pre-registered scalar hold-or-jump filter run causally on the CES history (Student-t measurement noise at the 10 ms bulk scatter) has a T_i median absolute error 0.906–0.920× persistence (8/8 split–population cells excluding 1); for V_{rot} its ratio is 0.94–1.00× (interval including 1 on 3/4 splits, also when fitted on the training files). The 10 ms scatter is consistent with independent Gaussian line-fit errors: the ratio of relative T_i to V_{rot} scatter is $\sqrt{2}$ on 8/8 cells (1.395–1.532), while the $\sqrt{T_i}$ scaling expected of photon statistics was not observed (slope 0.09–0.23 against 0.5); between labels the median-optimal point is the last value. The filter's MSE skill against the backbone is −0.06 to −0.52, because event rows stay late.

7. Limitations and conclusion

Limitations. The aggregate skill is a tail statistic; typical rows are not improved over PCHIP. The fast channels are standardized with whole-discharge statistics, an offline step, so a deployable variant is untested under this protocol; under an earlier protocol a training-statistics variant kept +0.22 to +0.30 over PCHIP (4/4) but was not scored against the causal GP. Two choices were made post hoc: the surrogate's form (chosen on the scored splits; its retention holds in the cut population only, inclusive 0.43–0.81) and the 1,000 km/s V_{rot} rule. The unit of replication is the shot; the reweighting reaches only 4–6% of missing V_{rot} rows. Everything is one chord, one device and one campaign.

Conclusion. For an estimate that cannot wait for the next CES exposure, the full-input causal estimator, with fast channels standardized per discharge, has T_i skill +0.113 [+0.085, +0.132] over the CES-only causal GP (cut); the fast-channel share is not isolated, and against an offline noise-aware smoother of both neighbours the skill interval includes zero. V_{rot} at this chord is estimated from its own history; its margin is established against persistence on every split and against the causal GP only in the pooled readout, and its 10 ms scatter is consistent with Gaussian line-fit errors. Code and evaluation artifacts: https://github.com/iseungsang01/ml-intern-revision.

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